Lecture 17: Principal Component Analysis

IEOR E4709: Statistic Analysis and Time Series
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1 Principal Component Analysis

We begin with a quick digression about *exploratory* versus *confirmatory* analysis. The goal of exploratory analysis is to reveal structures of the data that would remain concealed if data were represented naively. The goal of confirmatory analysis is to test hypotheses about the data. Our next focus is on principal component analysis, that is part of exploratory analysis.[1] Consider a $n \times p$ data matrix, $X = (x_{ij})$, where each observation $(x_{i1}, x_{i2}, \ldots, x_{ip})$ consists of p variables (e.g. we can think of each row of this matrix as consisting of the vector of DJ30 stock prices at time t_i). Principal component analysis (PCA) seeks a compact summary of the variation of the data matrix X. That is, PCA seeks a few linear combinations of the variables that summarize as much of the variation in the data as possible.

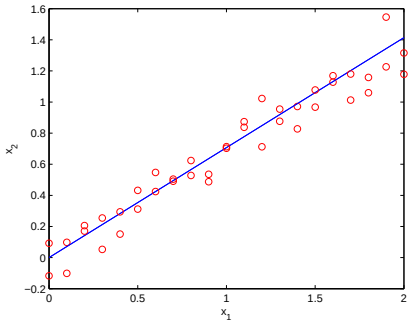


Figure 1: The new variable $\alpha = \frac{1}{\sqrt{2}}(x_1 + x_2)$ describes most of the variation in the data.

Denote by μ the mean vector, i.e., $\mu_j = \frac{1}{n}\sum_{i=1}^{n} x_{ij}$, for $j = 1, \ldots, p$. In general, we seek *principal components* $\alpha_i = (X - \mu)v_i$ such that, for $||v_i|| = 1$, the first principal component is the linear combination of the variables that has the largest variance. The second principal component is the linear combination of the variables that has the largest variance subject to being uncorrelated

[1] We will avoid using bold symbols for vector and matrices to avoid cluttered notation. It should be clear from the context if a quantity is a scalar, vector, or matrix.

with the first principal component, etc... Let us further assume that the mean μ of the columns of X is zero (if it not zero, then we subtract the mean from each column of the data matrix), hence $\alpha_i = Xv_i$. Recall that, given two zero-mean n-dimensional vectors x and y, their sample covariance is

$$cov(x, y) = \frac{1}{n-1} x^\top y.$$

In particular, the sample variance of the principal component α_i is

$$var(\alpha_i) = cov(\alpha_i, \alpha_i) = \frac{1}{n-1} \alpha_i^\top \alpha_i = \frac{1}{n-1} v_i^\top X^\top X v_i = v_i^\top \Sigma v_i,$$

where $\Sigma = \frac{1}{n-1} X^\top X$ is the sample covariance of the data. We want to choose v_1 so to maximize the variance of the first principal component subject to the constraint that $||v_1|| = 1$. This amounts to maximizing

$$\chi^2 := v_1^\top \Sigma v_1 - \lambda_1 (v_1^\top v_1 - 1),$$

where λ_1 is the Lagrangian multiplier. We then obtain that $\frac{\partial}{\partial v_1} \chi^2 = 0$ if and only if $2\Sigma v_1 - 2\lambda_1 v_1 = 0$, i.e., $\Sigma v_1 = \lambda_1 v_1$. The weight vector v_1 is called the principal component vector, and is the eigenvector of the covariance matrix Σ with the largest eigenvalue.[2] To see why v_1 must be the eigenvector associated with the largest eigenvalue, observe that $var(\alpha_1) = \frac{1}{n-1} \alpha_1^\top \alpha_1 = v_1^\top \Sigma v_1 = v_1^\top \lambda_1 v_1 = \lambda_1$,

The principal component $\alpha_2 = Xv_2$, i.e., the projection of the data matrix X onto the eigenvector associated with the second largest eigenvalue of Σ. It can be easily seen that α_2 is uncorrelated with α_1. This follows from the fact that the eigenvector v_2 is orthogonal to v_1 because Σ is a symmetric matrix and, for a real symmetric matrix, the eigenvectors corresponding to different eigenvalues are orthogonal. First, from the formula of the covariance matrix of two vectors, we obtain that

$$cov(\alpha_i, \alpha_j) = \frac{1}{n-1} \alpha_i^\top \alpha_j = \frac{1}{n-1} v_i^\top X^\top X v_j = v_i^\top \Sigma v_j.$$

Hence, $cov(\alpha_1, \alpha_2) = v_1^\top \Sigma v_2 = \lambda_2 v_1^\top v_2 = 0$. Hence, α_2 is the linear combination of the variables with the largest variance subject to being uncorrelated with the first principal component. Hence, the PCA yields uncorrelated principal components that are ordered according to decreasing variance λ_i's.

1.1 Definitions

More formally, the *principal component vectors* (aka principal component loadings or empirical orthogonal functions) v_i are the eigenvectors of the sample covariance matrix Σ, i.e., satisfying $\Sigma v_i = \lambda_i v_i$, where we use the convention that $\lambda_1 \geq \lambda_2 \ldots \geq \lambda_p \geq 0$. Notice that each eigenvector is of length p.[3] The *principal components* α_i (p component vectors of length n) are linear combinations of the columns of the data matrix X, whose weights in the linear combination are the components

[2] A real symmetric matrix Σ may have eigenvalues with algebraic multiplicity higher than 1, hence there may be multiple eigenvectors corresponding to the same eigenvalue. Then any linear combination of those vectors is also an eigenvector, with the same eigenvalue. For instance, if Σ is the identity matrix, then every vector is an eigenvector, and most PCA routines would return an essentially arbitrary collection of mutually perpendicular vectors. Typically, however, any arbitrarily small perturbation to Σ will break the degeneracy, i.e. it will guarantee that each eigenvalue is associated with only one eigenvector.

[3] The expection is if $n < p$, in which case there are only n distinct eigenvectors and eigenvalues.

of the eigenvector v_i: $\alpha_i = Xv_i$. The j-th element of the principal component α_i is called the *score* of the i-th principal component on the j-th observation.

1.2 Variance of Principal Components

The eigenvalue λ_i associated with the principal component vector v_i represents the variance of the i-th principal component, given by $var(\alpha_i) = \frac{1}{n-1}\alpha_i^\top \alpha_i = v_i^\top \Sigma v_i = v_i^\top \lambda_i v_i = \lambda_i$, using that $v_i^\top v_i = 1$. Divide the i-th eigenvalue λ_i by the sum of all eigenvalues

$$\frac{\lambda_i}{\sum_{i=1}^p \lambda_i} = \frac{\lambda_i}{tr(\Sigma)} = \frac{Var(i^{th} \text{ principal component})}{\text{Total variance}},$$

where $tr(\Sigma)$ is the trace of the matrix Σ, i.e. the sum of its diagonal entries that also coincide with the sum of its eigenvalues. This ratio is the proportion of total variation accounted for by the i-th principal component. Hence,

$$\frac{\sum_{i=1}^k \lambda_i}{\sum_{i=1}^p \lambda_i}$$

is the proportion of total variation accounted for by the first k principal components.

1.3 Property of maximum variance

No linear combination of the columns of X, Xu, with $||u|| = 1$, has a variance larger than λ_1, which is the variance of the first principal component α_1. If Xu is a linear combination of variables (with $||u|| = 1$) that is uncorrelated with the first k principal components $\alpha_1, \ldots, \alpha_k$, then the variance of Xu is maximized when Xu is the $(k+1)$-th principal component $\alpha_{k+1} = Xv_{k+1}$.